MINT CLASSICS MODEL CARS

Analysis Report

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PROJECT SCENARIO

Mint Classics Company, a retailer of classic model cars and other vehicles, is looking at **closing one of their storage facilities**.

To support a data-based business decision, they are looking for suggestions and recommendations for **reorganizing or reducing inventory**, while still maintaining timely service to their customers.

OBJECTIVES

1. Explore products currently in inventory and identify patterns.
2. Determine important factors that may influence inventory reorganization/reduction.
3. Provide analytic insights and data-driven recommendations.

SUMMARY

Mint Classics, a retailer of classic model cars and other collectible vehicles, operates in 7 offices across the globe. EMEA and NA are their largest customers, each taking roughly 40% of the market share. Their inventory is stored between four warehouses (North, East, South, West), each having 1-3 unique product lines. These facilities are not organized by territory, meaning all four are actively shipping items across the globe.

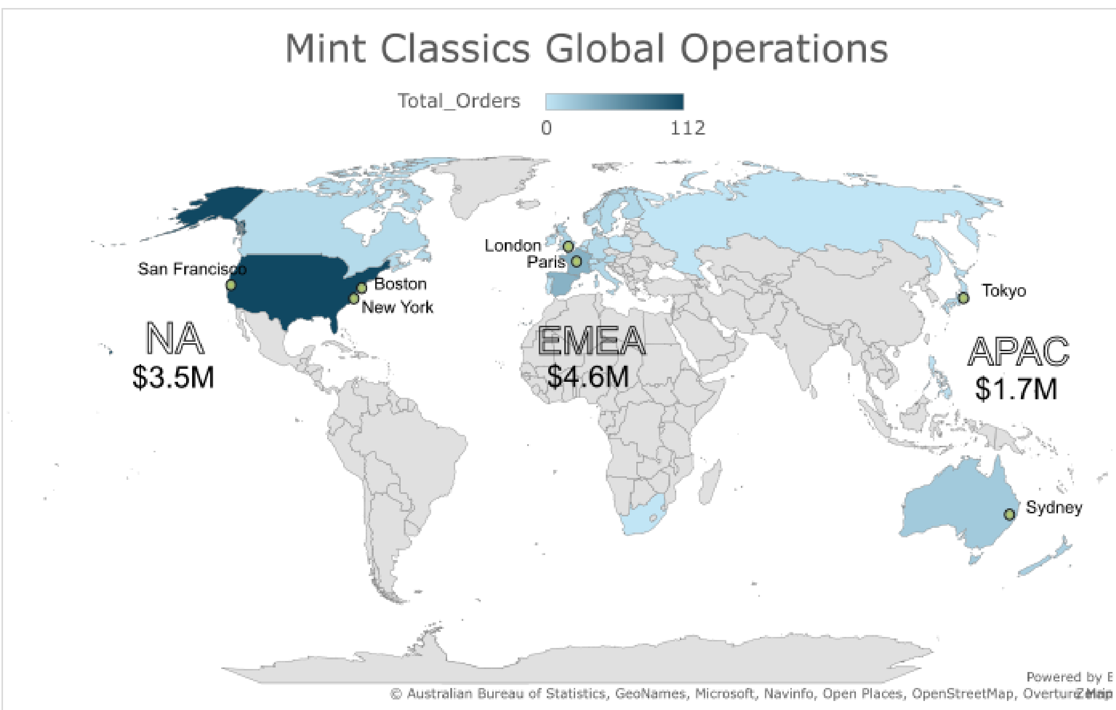
To optimize their storage capacity and align inventory with sales figures and objectives, I conducted a deep analysis of their business dynamics, identifying cold product stocks and low-sales product lines, to come up with a complete reorganization while considering their customers' voices.

STEP-BY-STEP APPROACH

1. Explore products currently in inventory:
   a. Imported mintclassics database to MySQL Workbench
   b. Create an Extended Entity-Relationship (EER) diagram and investigate inter and intra table relationships (foreign keys, primary keys)
   c. Data quality checkup using SQL:
      i. Check dates formatting and normalize to yyyy-mm-dd
      ii. Check missing or duplicated information and update tables if necessary

    d. Use SQL to retrieve information from multi-table relationships and provide relevant statistics
2. Determine important factors that may influence inventory reorganization/reduction:
    a. What is the role of storage facilities regarding global shipping? Are some warehouses more efficient than others?
    b. Where are items stored and if they were rearranged, could a warehouse be eliminated?
    c. How are inventory numbers related to sales figures? Do the inventory counts seem appropriate for each item?
    d. Are we storing items that are not moving? Are any items candidates for being dropped from the product line?
3. Formulate suggestions and recommendations for solving the business problem:
    a. Provide analytic insights and data-driven recommendations
    b. Create what-if scenarios
4. Final conclusions

RESULTS

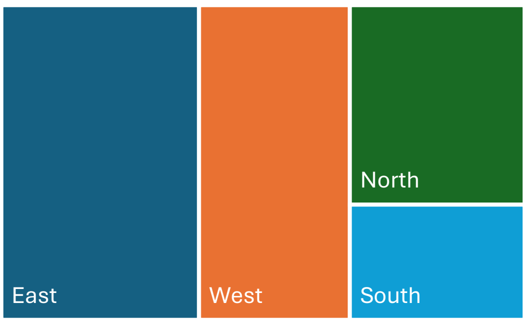


*Legends: **NA**: Canada, USA; **EMEA**: UK, Denmark, Belgium, Sweden, Finland, Norway, Germany, Spain, Italy, France, Austria, Switzerland; Non-active clients in: Russia, South Africa, Poland, Netherlands, Portugal; **APAC**: Australia, New Zealand, Japan, Singapore, Philippines.*

USA is the company's largest single-country customer, although the most profitable region is EMEA. There are 6 inactive customers in the region and most of them do not have a representative assigned; it might be beneficial to add another office or re-contact clients.

I analyzed how the storage facilities (their distribution, capacity, product lines, tied capital and revenue) are tied to global sales. Limitations should be pointed out, since there is no data on maintenance costs, location or shipping distance from warehouses to customers.

Relative size of warehouses



Units shipped across regions from each storage

	EMEA	NA	APAC	**Total**
North	10619	8996	5035	**24650**
East	18292	12111	5179	**35582**
West	9660	8755	4518	**22933**
South	11007	8275	3069	**22351**
Total	49578	38137	17801	105516

Warehouse	**Capacity**	**Product Lines**	**sold to stock ratio (%)**	**Tied Capital ($)**	**Historical Revenue ($)**	**Avg Revenue per unit ± SD**
North	72%	Motorcycles	15.5	3,565,714.18	1,113,502.03	87.32±35.8
		Planes	14.8	3,099,282.76	869,810.90	80.56±24.5
East	67%	Classic Cars	14.63	14,059,337.71	3,670,560.34	108.66±42.1
West	50%	Vintage Cars	15.16	5,704,259.82	1,674,618.33	78.19±33.5
South	75%	Trucks and Buses	22.5	2,139,329.56	968,566.87	92.74±26.0
		Ships	22	1,239,140.43	588,400.06	78.06±33.5
		Trains	13.7	727,251.77	175,030.77	66.02±16.6

The relative size of warehouses was calculated based on their current stock and the warehouse occupation percentage provided on the tables. Shipping volume originating from each storage is proportional to their current stock and the regional market size. Customers receive shipping from different warehouses in the same order, since most orders contain different product lines.

East warehouse holds 'Classic cars', which is the most profitable product line - with the highest revenue per unit sold. West is the second in size and revenue, mainly due to high volume of sales. The combination of product lines in North and South is clever; by doing so, they can match West warehouse total gains; some of their product lines have quite high revenue per unit. Surprisingly, South did great on historic income relative to its capacity, showing a good use of space and a dynamic inventory.

The table below shows the 'stock health' of each warehouse: a measurement that helps identifying dead-stored products. Alarmingly, ~40 to ~80% of items are overstocked, but just one item has dead sales. West and North warehouses have the highest amount of overstock items, compared to their size and total distinct products.

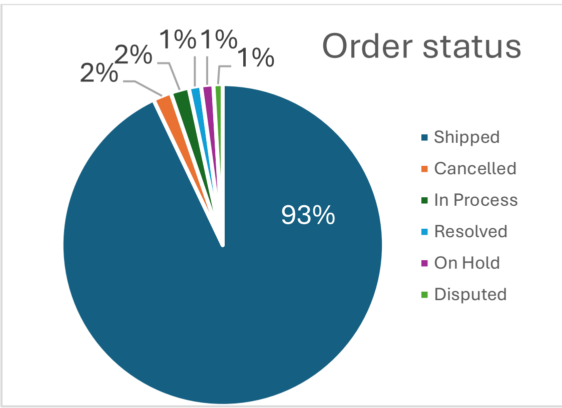
warehouse Name	total distinct products	severe understock%	normal balance %	severe overstock%	dead sales%
East	38	2.63	18.42	76.32	2.63
North	25	16	8	76	0
South	23	4.35	52.17	43.48	0
West	24	12.5	16.67	70.83	0

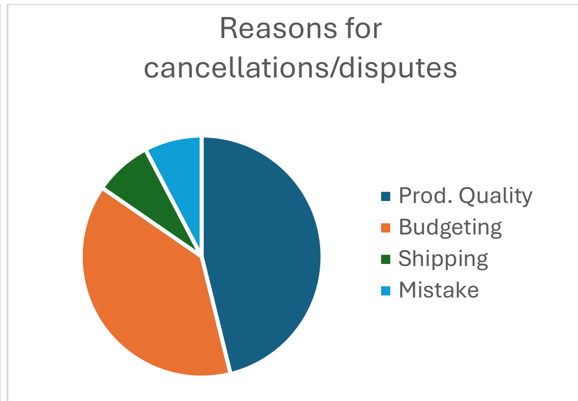
RANKING OF MOST AND LEAST SOLD ITEMS BY UNIT VOLUME

RANK	Stored In	Product Name	product Line	avg price	Sold (U)	stock (U)	Revenue generated
1 MOST SOLD	East	1992 Ferrari 360 Spider red	Classic Cars	33.87	1808	8347	$277K
2 MOST SOLD	West	1937 Lincoln Berline	Vintage Cars	20.55	1111	8693	$103K
3 MOST SOLD	North	American Airlines: MD-11S	Planes	14.81	1085	8820	$72K
4 MOST SOLD	West	1941 Chevrolet Special Deluxe Cabriolet	Vintage Cars	21.17	1076	2378	$103K
5 MOST SOLD	West	1930 Buick Marquette Phaeton	Vintage Cars	8.73	1074	7062	$42K
5 LEAST SOLD	West	1911 Ford Town Car	Vintage Cars	12.11	832	540	$45K
4 LEAST SOLD	West	1936 Mercedes Benz 500k Roadster	Vintage Cars	8.21	824	2081	$30K
3 LEAST SOLD	East	1970 Chevy Chevelle SS 454	Classic Cars	13.96	803	1005	$53K
2 LEAST SOLD	East	1957 Ford Thunderbird	Classic Cars	14.25	767	3209	$50K
1 LEAST SOLD	East	1985 Toyota Supra	Classic Cars	NA	0	7733	$0

Next, I proceeded with the identification of the most and least popular items in the inventory, as well as inventory movements related to sales. The above table shows that some of the least-sold items gave relatively high revenue (highlighted in yellow). It is clear that 1985 Toyota Supra model should be discontinued since no units have been sold from this product.

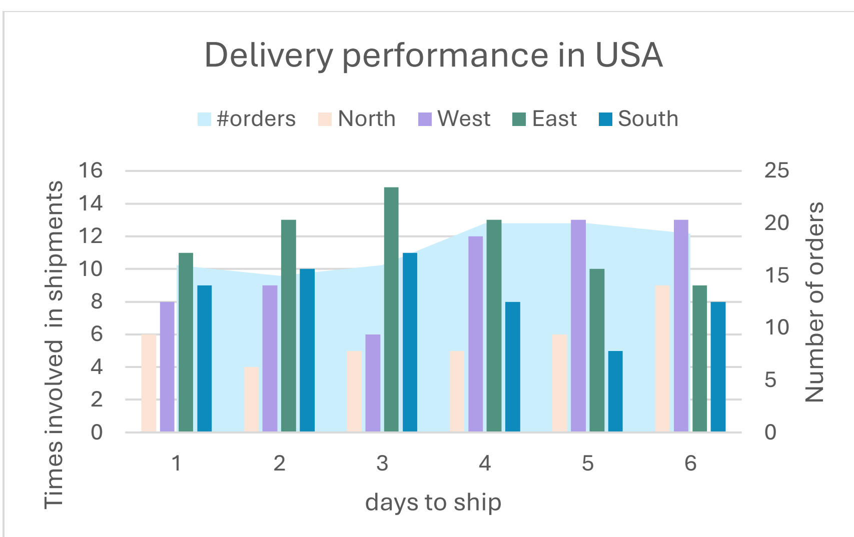
A less orthodox approach to measure warehouse performance is to follow the comments in the orders table, which might uncover shipping or storage issues. According to a background analysis, only 2% of orders are cancelled; reasons are assorted but mostly it is budget or credit limit. Product quality issues are significant as well, but they are random customer preferences rather than defects on the product line (data not shown here).





A third important factor in storage performance is the number of late deliveries. We know that all warehouses ship to all countries, and variables in international shipping are too many to consider. To make the analysis simpler, I focused on the orders shipped to USA and counted the times each warehouse was involved in deliveries in a range of days, to observe any pattern on possible delays.

The chart below shows that most purchases are shipped between 4-5 days after ordering. Although the pattern is very subtle, West warehouse participates more in later deliveries than early ones - opposite for North and East. We could say that there is no reason for this warehouse to perform better in international shipping either. This information might just be complimentary to more relevant aspects of warehouse performance to recommend directional actions.



INSIGHTS AND CONCLUSIONS

Key takeaways

- No evident link was found between warehouses and orders cancellations or disputes.
- There is no reason to reorganize warehouses by regions, since there are no correlations on most-sold and least-sold product lines with shipping and total revenue.
- East warehouse is the number one in ranking capacity and provides the higher revenue, even though its stock is slow-moving; South is the smallest in size and revenue, working almost at its full capacity, but has a healthy stock with items moving frequently; and West is the second largest facility, working at half capacity and probably experiencing delays in shipping.
- The 3 largest warehouses are alarmingly overstocked in low-selling products, while the most sold products need immediate restock. There might be an issue with repository or suppliers.
- Delivery performance is similar between warehouses; however, the average number of days from ordering to shipping could be improved.
- Taking into consideration the high volatility on revenue per unit shown in the charts above, dropping items from the product line or even calculating revenue per warehouse based just on volume sales is not appropriate; each model must be analyzed separately.

SOLUTIONS

Following the SQL analysis and conclusions, I propose these actions:

- There are two possible options for closing a storage facility:
  1. Closing the South warehouse and distributing its product lines as follows: Trains and Ships to North and Trucks and Buses to West. This decision is based solely on stock volume and capacity on the remaining storages, since this facility is overall in good shape.
  2. Close West warehouse and move items to East. West is a large facility, therefore more expensive, and is working at 50% of capacity. Moreover, it is associated with slow deliveries and severe overstock.
     - However, East warehouse cannot absorb the entire West inventory, because its capacity would be saturated.
     - Splitting a product line in several warehouses is not a good idea either; it can introduce complications with suppliers and post-sales operations
- To overcome these issues, I suggest:
  - Reducing 20% of each overstock volume in East and West warehouses and eliminating product S18_3233 from the product line before the closure of West.
  - This could be achieved by sales events or promotions.

What If simulations: an overlook of rearrangements on selected warehouse and its impact on tied capital and revenue.

Capac.	Simulated Warehouse	New Product Lines	Simulated Stock Qty	Simulated Tied Capital ($)	Simulated Shipped Revenue ($)
91.60%	North	Motorcycles, Planes, Trucks and Buses	167,539	8,804,326.50	2,866,661.74
67.40%	West	Vintage Cars, Ships, Trains	168,409	7,670,652.02	2,374,832.27
67.00%	East	Classic Cars	219,183	14,059,337.71	3,623,600.63

1- reassignment of South product lines to North and West warehouses

Capac.	Simulated Warehouse	New Product Lines	Simulated Stock Qty	Simulated Tied Capital ($)	Simulated Shipped Revenue ($)
105.20%	East	Classic cars, Vintage Cars	344,063	19,763,597.53	5,266,773.12
75.00%	South	Trucks and buses, Ships, Trains	79,380	4,105,721.76	1,680,664.12
72.00%	North	Planes, Motorcycles	131,688	6,664,996.94	1,917,657.40

2- reassignment of West product lines to East warehouse. We can see the 105% capacity overflow.

Capac.	Simulated Warehouse	New Product Lines	Simulated Stock Qty	Simulated Tied Capital ($)	Simulated Shipped Revenue ($)
83.2%	East	Classic cars, Vintage Cars	272,116	15,586,733.87	5,266,773.12
75.00%	South	Trucks and buses, Ships, Trains	79,380	4,105,721.76	1,680,664.12
72.00%	North	Planes, Motorcycles	131,688	6,664,996.94	1,917,657.40

3- Reducing 20% of the original East and West stock in selected items before merging. Note: I did not calculate a possible revenue gained by this 20% reduction; failed to predict the sold price and total sales per item in case of a reduced-price sales event. Note: restock of items is not taken into consideration

SQL scripts repository used for this report is organized as follows:

1. Data cleared: critical step to ensure correct formatting of dates, and updated Japan into APAC region
2. Background analysis: queries used to describe the overall business organization (regional sales and price gaps, customers demographics and active accounts, flagged customers)
3. Cancellations and customer satisfaction: queries to answer if comments on orders table could help in determining warehouse connections to cancellations and disputes
4. Delivery performance: queries used to find out the average days to ship per country or region and map the warehouse-to-region shipping.
5. Inventory vs sales: queries for calculating the most and least sold items, average revenue per unit and total revenue of product lines.
6. Stock movement: SQL queries for observing the stock health per warehouse (how many items were sold compared to stored) and recalculating the inventory reduction and restock

7. Warehouse rearrangement: SQL queries to determine capacity ranks, product diversity, original distribution of product lines and what if scenarios.

Supplementary tables

A) Details on stock reduction per item

warehouse Name	product Code	Product Name	product Line	total units sold lifetime	current stock	target reduction units	remaining stock	stock health status
East	S18_3233	1985 Toyota Supra	Classic Cars	0	7733	7733	0	Absolute Dead
East	S18_4933	1957 Ford Thunderbird	Classic Cars	701	3209	641	2568	Severely Overstock
East	S18_2870	1999 Indy 500 Monte Carlo SS	Classic Cars	777	8164	1632	6532	Severely Overstock
East	S18_1589	1965 Aston Martin DB5	Classic Cars	831	9042	1808	7234	Severely Overstock
East	S24_4048	1992 Porsche Cayenne Turbo Silver	Classic Cars	844	6582	1316	5266	Severely Overstock
East	S24_3432	2002 Chevy Corvette	Classic Cars	851	9446	1889	7557	Severely Overstock
East	S24_1628	1966 Shelby Cobra 427 S/C	Classic Cars	851	8197	1639	6558	Severely Overstock
East	S24_3191	1969 Chevrolet Camaro Z28	Classic Cars	857	4695	939	3756	Severely Overstock
East	S24_2972	1982 Lamborghini Diablo	Classic Cars	872	7723	1544	6179	Severely Overstock
East	S12_3990	1970 Plymouth Hemi Cuda	Classic Cars	875	5663	1132	4531	Severely Overstock
East	S18_1984	1995 Honda Civic	Classic Cars	884	9772	1954	7818	Severely Overstock
East	S12_3380	1968 Dodge Charger	Classic Cars	903	9123	1824	7299	Severely Overstock
East	S10_4962	1962 LanciaA Delta 16V	Classic Cars	906	6791	1358	5433	Severely Overstock
East	S18_3482	1976 Ford Gran Torino	Classic Cars	915	9127	1825	7302	Severely Overstock
East	S18_3685	1948 Porsche Type 356 Roadster	Classic Cars	917	8990	1798	7192	Severely Overstock
East	S10_4757	1972 Alfa Romeo GTA	Classic Cars	917	3252	650	2602	Severely Overstock

East	S18_1129	1993 Mazda RX-7	Classic Cars	921	3975	795	3180	Severely Overstock
East	S18_4027	1970 Triumph Spitfire	Classic Cars	922	5545	1109	4436	Severely Overstock
East	S12_3148	1969 Corvair Monza	Classic Cars	933	6906	1381	5525	Severely Overstock
East	S24_4620	1961 Chevrolet Impala	Classic Cars	941	7869	1573	6296	Severely Overstock
East	S24_3371	1971 Alpine Renault 1600s	Classic Cars	945	7995	1599	6396	Severely Overstock
East	S18_1889	1948 Porsche 356-A Roadster	Classic Cars	949	8826	1765	7061	Severely Overstock
East	S12_4675	1969 Dodge Charger	Classic Cars	951	7323	1464	5859	Severely Overstock
East	S10_1949	1952 Alpine Renault 1300	Classic Cars	961	7305	1461	5844	Severely Overstock
East	S12_1108	2001 Ferrari Enzo	Classic Cars	973	3619	723	2896	Severely Overstock
East	S24_1444	1970 Dodge Coronet	Classic Cars	976	4074	814	3260	Severely Overstock
East	S18_2238	1998 Chrysler Plymouth Prowler	Classic Cars	986	4724	944	3780	Severely Overstock
East	S700_2824	1982 Camaro Z28	Classic Cars	997	6934	1386	5548	Severely Overstock
East	S24_3856	1956 Porsche 356A Coupe	Classic Cars	1013	6600	1320	5280	Severely Overstock
East	S18_3232	1992 Ferrari 360 Spider red	Classic Cars	1748	8347	1669	6678	Severely Overstock
West	S24_3816	1940 Ford Delivery Sedan	Vintage Cars	771	6621	1324	5297	Severely Overstock
West	S18_3140	1903 Ford Model A	Vintage Cars	778	3913	782	3131	Severely Overstock
West	S24_3420	1937 Horch 930V Limousine	Vintage Cars	806	2902	580	2322	Severely Overstock
West	S18_1749	1917 Grand Touring Sedan	Vintage Cars	842	2724	544	2180	Severely Overstock
West	S18_4409	1932 Alfa Romeo 8C2300 Spider Sport	Vintage Cars	860	6553	1310	5243	Severely Overstock
West	S24_3151	1912 Ford Model T	Vintage Cars	881	9173	1834	7339	Severely Overstock

		Delivery Wagon						
West	S18_4522	1904 Buick Runabout	Vintage Cars	892	8290	1658	6632	Severely Overstock
West	S18_3136	18th Century Vintage Horse Carriage	Vintage Cars	907	5992	1198	4794	Severely Overstock
West	S50_1341	1930 Buick Marquette Phaeton	Vintage Cars	935	7062	1412	5650	Severely Overstock
West	S24_1937	1939 Chevrolet Deluxe Coupe	Vintage Cars	937	7332	1466	5866	Severely Overstock
West	S18_2325	1932 Model A Ford J-Coupe	Vintage Cars	957	9354	1870	7484	Severely Overstock
West	S18_1367	1936 Mercedes-Benz 500K Special Roadster	Vintage Cars	960	8635	1727	6908	Severely Overstock
West	S24_4258	1936 Chrysler Airflow	Vintage Cars	983	4710	942	3768	Severely Overstock
West	S18_2957	1934 Ford V8 Coupe	Vintage Cars	985	5649	1129	4520	Severely Overstock
West	S18_3320	1917 Maxwell Touring Car	Vintage Cars	992	7913	1582	6331	Severely Overstock
West	S18_4668	1939 Cadillac Limousine	Vintage Cars	995	6645	1329	5316	Severely Overstock
West	S18_2949	1913 Ford Model T Speedster	Vintage Cars	1038	4189	837	3352	Severely Overstock
West	S18_1342	1937 Lincoln Berline	Vintage Cars	1111	8693	1738	6955	Severely Overstock